

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper B)

July 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- Lumpsum in advance:
 - Funds that are expected to meet the cost of the benefit can be set up as soon the benefit promises are made.
 - The funds are designed to meet all future benefit outgo
 - In this method of funding on which payments are made as early as possible
 - The fund may end being too low or too high depending on the actual experience of the scheme
- Terminal Funding
 - The funds expected to meet the cost provide the series of benefit payment can be made as soon as first payment in made
 - The payment is the capital sum sufficient to meet all the future payment of the benefits
 - When the terminal funding is done the fund may not exist under the scheme like in case of lump sum payment on retirement
- Regular contributions
 - The funds are gradually built up to a level expected to meet the future benefit pay-out
 - Example contributions made during the working life of the member/employee to provide pension on retirement
 - The level of funding may vary from year-to-year based on the valuation of the scheme
 - It is usually the preferred method of funding the scheme
- Just-in time funding
 - The funds that are expected to be sufficient to meet the cost of the benefit can be set up as soon as the risk arises in relation to future financing
 - It differs from terminal funding in the sense that the trigger point is some external event which jeopardises the security of the fund
 - Example: When company solvency is threatened
- Smoothed PAYG
 - Funds are set up to smooth the cost under the PAYG approach to allow for the difference in the timing of contributions and benefits
 - The idea is to smooth income and outgo over time by maintaining a working balance
 - These systems are often operated by governments
 - Example: In some years when the tax income is higher opposed to contribution the working balance will increase and in other years it will decrease due to payments more than the income.

[Max 10]

ii)

- The cost of insuring the scheme may be higher than if self-managed
- The higher cost could be on account allowance of
 - Profit and expenses of the insurer
 - The solvency requirement for the insurer
 - Margins for uncertainty in experience
- If scheme has additional assets, then it may be refunded to the company

- On the other hand, if there is a shortfall it will lead to contribution requirement from the company at time of the scheme transfer
- There could be opportunity cost to the company in case there is shortfall
- The risks related to the managing the scheme will get transferred to insurer which include
 - Longevity risk
 - Investment Risk
 - Operational risks
 - Expense Risks
 - Liquidity Risk
- On the other hands it means that it will not be able to participate in the upsides which include
 - Investment return
 - Positive mortality experience
- Reputation risk to the company as result of failure of the insurer to provide the benefits

[Max 6]

iii)

- The past experience may not be reflective of the future mortality experience
- The data of the scheme may not be credible however that will depend on the size of the scheme
- The data may be sparse at older ages due to fewer number of members reaching that age
- Or at younger ages due to lower number of deaths
- The mix of lives may have changed over the period of time
- Any data issues or errors may be present in the experience investigation and therefore some adjustments may be allowed for the known or unknown errors
- There could be random fluctuations in the past data
- The one-off events in the past mortality experience like pandemic which may not be expected in near future
- Allowance for any IBNR claims
- The future mortality improvement may not be in line with what has been experienced in the past
- It is important to use assumptions appropriate to the purpose and type of valuation
- Since the valuation is for deriving the transfer value, margin may be required to be built in the mortality experience and mortality improvement in the future
- Otherwise, it could lead to underestimate of the mortality/longevity risk
- and hence lower transfer value which will be detrimental to the interest of the insurer
- May need to ensure consistency with the assumptions used to value similar schemes
- or any information of the past assumptions used by the manufacturer to value the liabilities
- May lead to non-compliance of professional guidance issued by actuarial professional

[Max 7]

iv)

- The quantum of investment in such projects could be large
- The transfer amount will free up capital for the Highway Authority which can be used to other road development purposes
 - This will help in increasing the road capacity of the country/state and aid economic development
 - Or generation of employment
 - Or can be used to pay off debt which may have been taken to construct the project
- The Authority may want to transfer post construction financial and non-financial risks to the private entity

- Private sector is believed to be more efficient, cost conscious and technology driven
- Higher private sector salaries and incentives that attract effective management personnel
- Help to create investment avenues for players with long term liabilities like provider of pension leading to win - win situation
- Absence of cumbersome regulations/protocols comparatively for the private player

[1/2 Marks each, Max 4]

v)

- Demand uncertainty and toll levels uncertainty can lead to inaccuracies in the estimation of the revenues which are supposed to cover operation and maintenance
- In fact, these risks are substantial due to the lack of credible forecasting methodologies
- Lack of credible data to estimate the parameters to be used in forecasting the revenue stream
- In addition, if the volume does only pick up in the later years and not to the extent forecasted in the earlier years - it will have an impact on the IRR to the investor
- Higher cost of general repair and maintenance of the roads due to inflation turning out to be higher than expected
- Any other convenient mode of transport which may spring up in future like bullet train
- Or change in the travelling habits of the population like preferring air travel
- Or any other parallel roads which may come up in the future leading to increase in the competition and lower revenue
- The change in the taxation policy impacting the net revenue stream and net returns
- The change in the government policies with respect to determining the tariffs and magnitude of increases in tolls.
- The risk of change in the policy where the road is nationalized later by the local government
- Force majeure risks are the risks related to natural catastrophes such as earthquakes or floods will impact revenue
- The repair and maintenance cost due to natural calamity destroying certain section of roads
- Travelling may be restricted by government like due to pandemic
- Traffic volumes are very sensitive to income and economic growth which may be on a lower end.
- Commercial traffic is usually more remunerative when compared to passenger traffic, it remains exposed to economic cycles.
- Increase in oil prices leading to fall in the travelling demand
- The leakages or internal theft at the toll plaza
- However, this will depend on whether the toll collection is automated or manual
- The cost in updating any toll collection mechanisms like eTag
- The new toll payment systems may not be accepted by the users.
- The operator will have to depend on the local authority/government in enforcement of the new system and the application of penalty on the users
- Also provide police assistance when required for traffic regulation or removal of trespassers
- The extent of convenience like rest rooms, restaurants etc on the toll road will also impact the volume
- The company does not have any experience in managing such project
- In case the maintenance is outsourced to third party then it could lead to additional risks
 - Counter-party Risk
 - Legal Risk
 - Poor quality of service standards
 - Conflicts in case of fines and penalty imposed by local government road agency
- In the event of poor maintenance, attractiveness of the road may diminish making commuters shift to alternative stretches,
- The quantum of investment may be large and therefore exposes the insurer to concentration risk

[Max 13]

vi) The following could be the reasons for the scheme having higher proportion of members opting for lump sum as compared to other schemes:

- The comparatively low amount of pension on retirement may lead to members preferring cash
- The lower level of pension is likely due to low salary at retirement or low past service
- One would expect that since it is a manufacturing company most of the employees are likely to have higher tenure however will belong to low salary bracket
- The members are unlikely to be equipped to self-manage the investment and generate high return on assets and therefore this cannot be the reason.
- Health of members will play an important role in higher lumpsum take up proportion
- The retirement age of the manufacturing company could be high meaning that members could suffer from old age ailments
- Post retirement they will lose access to the group health insurance scheme of the company
- Low number of members may have health insurance policies as they may belong to low socio-economic class
- Health cover becomes difficult to get at an advanced age
- The exposure to harmful substances in the manufacturing company over a long period of time may lead to poor health of employees at older ages
- Any ongoing pandemic will increase health cost burden of those already in poor health
- Or they may prefer to take impaired annuities from third party if majority of them are in poor health
- Significant proportion of employees may prefer cash for paying mortgages or/and financing education for children
- Particularly there has been a significant increase in the inflation of some of the basic items like education, healthcare, and housing
- The salary increments of particularly white-collar employees like those employed by manufacturer have not been able to keep pace with the same
- There has been a significant increase in the home values in cities where the manufacturing factories are based
- Or significant benefits provided by the manufacturer during the tenure like
 - ESOPS
 - Annual bonuses
- This would mean that the members will have more confidence to take lump sum and spend the money
- The lack of financial awareness or guidance received due to lower level of education or financial access
- Example taking lumpsum exposes member to longevity risk
- Favourable tax treatment if the cash value is below certain threshold
- Any errors in calculating the cash value for the above scheme leading to higher cash values paid out leading to more people opting for lump sum
- The members of the scheme do not have financial confidence in the insurer to provide the benefits

[Max 10]

[50 Marks]

Solution 2:

i) Benefits:

- The image of insurer may increase for promoting wellness among insured population.
- May be joining with competitors on these schemes.
- The claims frequency may be reduced because of healthy life style of policyholders.
- The severity of claims may also decrease for indemnity type of products.
- The insurer will get more data on healthy habits of the insured, which may help in more customised underwriting and pricing.
- May be looked as a new age insurer.

Drawbacks:

- The price of health insurance product may be unaffordable or the insurer may not offer the insurance coverage to certain segments of population who doesn't have certain healthy habits.
- This may create negative image if the insurance coverage for everybody is motto of regulator or the government.
- If the gadgets are not working and if any fraud happens, the company's reputation may be damaged.
- Focus of regulator may increase on such complaints.
- Redemption of points earned over a period of 3yrs may require holding separate reserve and interest rate will play an important role
- Administrative cost may increase
- If the criteria are not transparent, there may be increased customer complaints.

[Max 5]

ii) Risks from the proposals:

- Data related risks:
 - Major risk is the lack of data on wellness related activities of policyholders and further break-up of such data
 - Difficult to get data on likely proportion of policyholders who will opt for such scheme.
 - The risk of more number of policyholders who will be eligible for redemption than estimated leading to more discount
 - There could be risk that the fitness conscious behaviour of the policyholder would not have resulted in improvement of claim experience under health insurance product.
 - Willingness of existing customers to participate in wellness program and willingness to adopt technology mayn't be as expected
- Redemption rate risk:
 - The rate at which the policyholders would be redeeming at various points in time would change.
 - This change may lead to risk in interest rate assumption used for arriving at the discount rate.
 - The risks mentioned above will become more onerous if the redemption formula is not reviewable.
- Operational risk:
 - Errors in data capture by gadgets or, fraud.
 - The above risk will further aggravate if the gadgets are of substandard and there is no control on what type of gadgets the policyholders are using.
 - Scope for change in business mix i.e., more young policyholders than middle and old age people
 - The gadgets mayn't be giving accurate information
 - Lack of expertise in administration of such schemes
 - The third party mayn't be successful as expected
 - The main risk that the framework for allotment of points based on health related activities may be incorrect. There could be improvements in gadgets and the redemption formula may require change.
 - The classification of risk i.e., based on number of points earned mayn't be a correct classification

- Legal risk:
 - There could be litigation risks if the data is leaked or shared by the third party vendor to some other party
 - There could be a change in regulatory provisions regarding data sharing and data privacy of policyholders
- Expense risk:
 - There could be a risk that the expenses assumed in availing this service from third party vendor would have raised.
- Competition risk:
 - There could be many competitors offering such schemes at more attractive rates.

[8]

iii) Mitigation of risks:

- Diversification of service providers:
- Multiple third party service providers may be engaged to avoid risk of dependence on single operator
- Instead of offering this feature to all policyholders it may be restricted to certain segment only.
- It may be offered to standard lives only.
- Reinsurers help may be taken
- In house development of such services may be explored
- The data collection, ownership may be made in-house rather than storing at third-party data centre
- Frequent checking of gadgets to check the accuracy of results
- Review of relationship between activities and points earned
- The activities done in recognised wellness centres may only be considered
- It may be offered only to certain age group of policyholders and may be to certain lines of business only (only for retail or individual policyholders)
- Stipulation of using certain gadgets or apps to avoid fraud
- Regular monitoring of claims experience whether the wellness has resulted in improvement in claims.

[5]

iv) Various steps:

- Arriving at the expected savings in claims cost:
 - The expected claims cost without wellness minus expected claims cost with wellness scheme
 - This is mainly from reduced frequency and reduced severity of claims due to regular physical activities.
 - This need to be examined by age-wise.
- Linking calories burnt, points earned and discount in premium:
 - The data on hours spent on physical activities and the calories burnt shall be collected.
 - Such data shall be on age-based.
 - If no public data is available, data from established gym or yoga centres may be obtained.
 - The relationship between calories burnt and expected improvement in rate of hospitalisation need to be framed.
 - As per this framework, points shall be allocated age-wise.
 - A further break-up may be made, if data available, as per the state of health of policyholder like a diabetic policyholder or policyholder with hyper tension.
 - The expected points earned by policyholders of different ages shall be arrived.

- Typically, the young age population may be earning more points.
- The more points, the likely benefits in terms of health related claims for middle and higher age population could be more.
- Based on these expected points, the expected cost with various percentages of discount in premium need to be arrived.
- As the young age policyholders likely to burn more calories, they are likely to earn more discount. However, the improvement in health related claims mayn't be significant as the frequency of health claims young population is less.
- However, the incentive at old age should not be significantly lower than young age.
- If it is negligible, there will not be incentive for middle aged or old age people.
- Hence, there could be a need to smoothen the discount percentages
- Redemption criteria of points over 3yrs period:
 - The expected time of redemption need to be arrived.
 - The interest rate that may be earned before redemption need to decided.
- Other costs that may need to be considered:
 - The cost of gadget and any payment to app developer need to be considered.
 - Any other fixed cost to be considered.

[9]

v) Aims of Regulation:

- Promote efficient and orderly markets
- Correct market inefficiencies
- Protect customers of insurance companies
- Maintain confidence in the insurance industry
- Help reduce financial crime

[3]

Expected Common area for both the proposals:

- May restrict the sharing of health data of policyholders
- May impose conditions on storage of data
- May impose conditions on terms and conditions with third party service provider
- May exclude or include certain terms and conditions in the policy document for offering this type of benefit
- May keep some limit on regular annuity amount
- May keep some restrictions on underwriting of customers
- May prescribe minimum discount to be offered on premium
- May restrict the wellness benefits for users of a recognised centres only
- May prescribe disclosure of certain facts about wellness benefits to prospective customers
- May prescribe different capital standards for health benefit and annuity benefits
- May prescribe different investment restrictions.
- May prescribe the disclosure of basis for framework for earning point and converting them to discount in premium
- May need to disclose the number of policyholders availed the discount and amount of discount.

[5]

[Max 7]

vi) Sources of data:

Proposal 1:

- As this is a new feature, internal data mayn't be useful

- Any data available from public sources like association of gyms or yoga centres may be useful in arriving at the age-wise and time spent
- Any reinsurers data in this regard may be useful
- Any data from global insurance market if such schemes are available in those jurisdictions
- Data on the link between calories burnt and improvement in health

Proposal 2:

- As the insurer is offering health insurance products also, any internal data may be used.
- If not relevant, any public information like Health insurance portal of government may be examined.
- Reinsurer's data may be taken
- For survival probability, the standard mortality table may be used
- If internal data on mortality experience of annuitant population is there, that may be used.
- For interest rate assumption, the data on interest rate movement either internally or externally

[2]

Assumptions required:

Proposal 1:

- Proportion of policyholders likely to redeem in various years.
- This depends on the likely points earned and the persistency rate of business
- The interest rate
- The discount proposed for each segment
- If the discount proposed varies by age-group or any other classification the expected proportion in such segments is required.
- The administrative cost
- The cost of gadgets, if they are supplied by insurer.

Proposal 2:

- Rate of hospitalisation
- Annuitants mortality rate
- Expenses assumption
- Investment return
- Mortality improvement factor

[4]

[6]

vii) Principles of Investment:

The principles of investment are:

The investments selected shall be appropriate to

- Nature
- Term
- Currency and
- Uncertainty

Of the liabilities and the company's appetite for risk

[2]

Subject to above the investments selected so as to maximise the overall return on assets.

Proposal 1:

- The benefit is fixed in nature as the amount is a proportion of premium
- The aggregate benefit amount may vary based on age mix of the portfolio.
- Thus there is uncertainty on the amount of benefit.
- The accumulation of points is for 3yrs but the earning of benefit is throughout premium payment term
- There is an uncertainty in timing of benefit as the redemption can happen anytime in 3yrs (though it coincides with premium payment due date)

Proposal 2:

- The benefit is fixed in nature as the amount one annual annuity.
- The aggregate benefit amount may vary based on age mix of the portfolio as the annuity amount varies by age.
- Thus there is uncertainty on the amount of benefit.
- The uncertainty in timing of benefit payment is linked to hospitalisation.
- With the rate of hospitalisation, the timing can be modelled. [2]

Suitability of money market instruments:

- They are highly suitable for short-term liabilities or for liabilities where there is high uncertainty.
- The return on money market instruments are lower compared to equity or bonds
- The duration of liability in respect of second option is relatively longer compared to option 1, the money market instruments mayn't be suitable for option2.
- As the uncertainty discussed above could reasonably be modelled, the money market instruments may not give the maximum overall return.

Bond instruments:

- The government bonds are highly secure and liquid.
- The tenor of these securities are usually available at different durations.
- Investment in these instruments may be suitable.
- However, the return earned may be lower than equities in longer term.
- As the benefits are not linked to inflation, government bonds may be suitable.
- The corporate bonds also are suitable subject to the acceptance of risk characteristics of these bonds.

Equity:

- The investment in equity instruments are risky but at the same time provide higher return over a longer period.
- They provide the inflation related returns.
- As the benefits are fixed in nature, this mayn't be major consideration for investment.

Other factors to be considered:

- Regulatory constraints on the limits of exposure, limits on investments under various instruments.
- The level of free assets with the company
- The risk appetite of the company.
- The liability under these benefits mayn't be significant portion of the overall liability of the company.

- Therefore, subject to regulatory constraints and risk appetite, the investment strategy of these liabilities mayn't be significant when compared with overall investment strategy.

[6]

[10]

[50 Marks]
